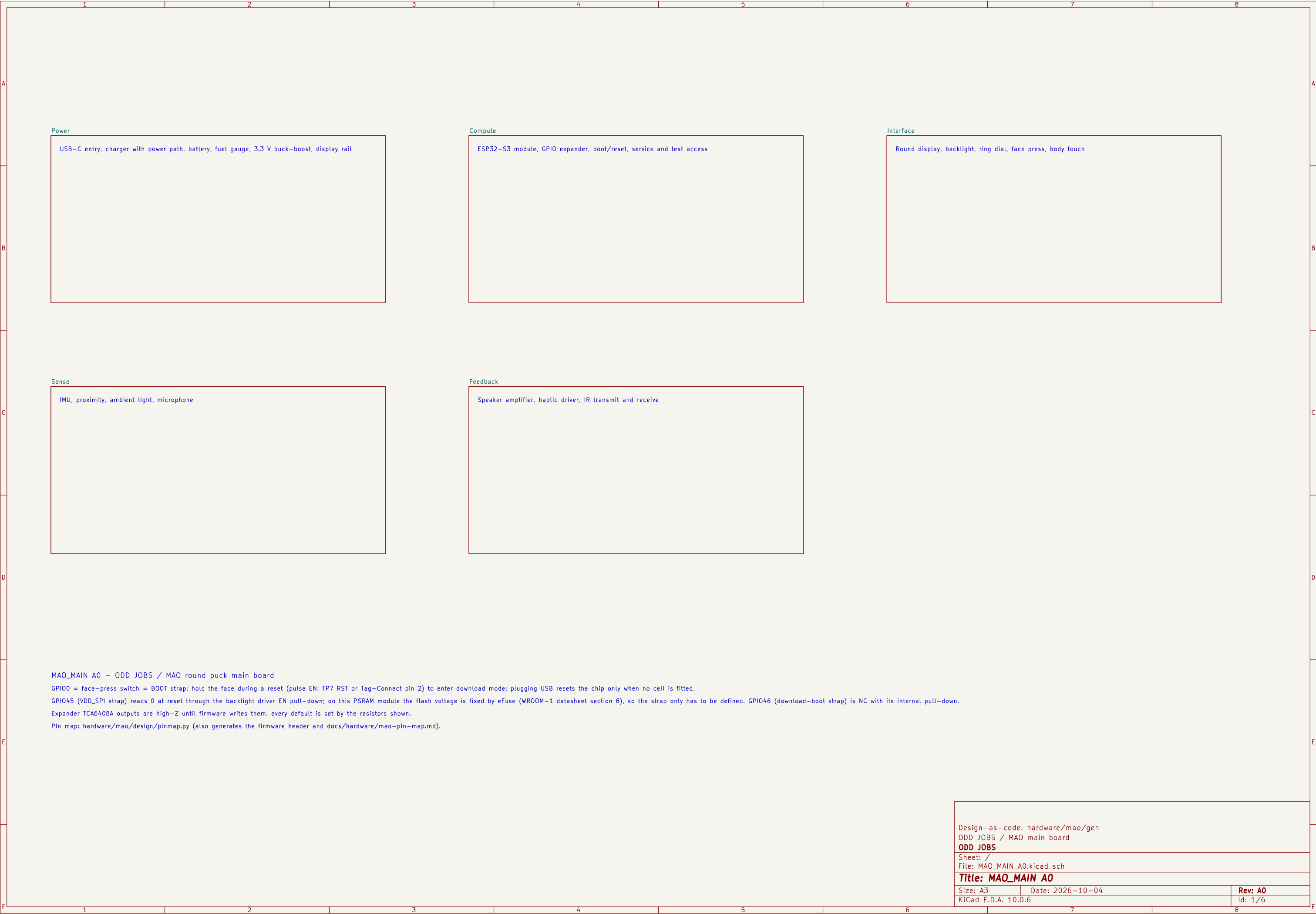
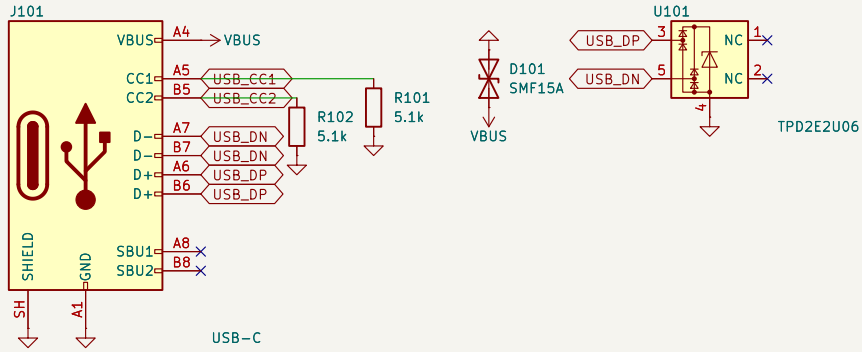




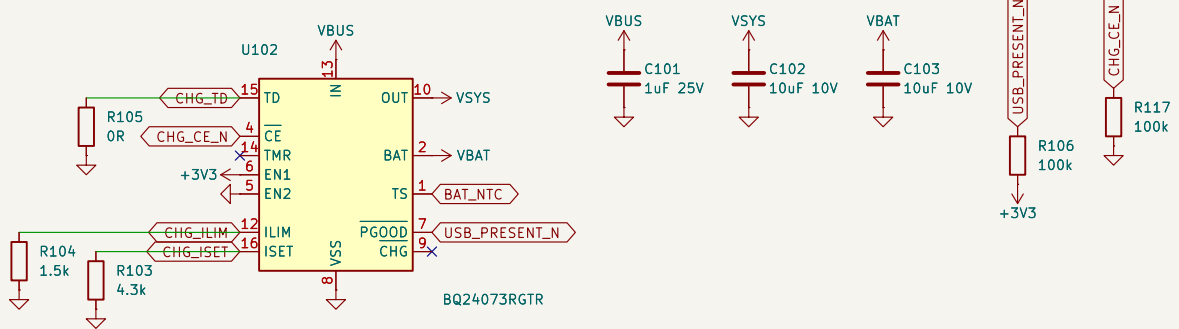
POWER

USB-C ENTRY



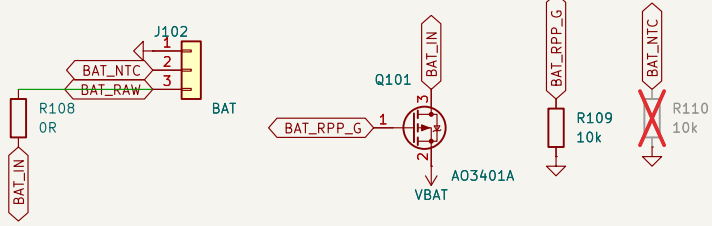
Sink only; 5.1k Rd on CC1/CC2 (ODD JOBS 25). ESD at the connector (119); TPD2E2U06 on D+/D-, SMF15A on VBUS. Shell to GND (plastic enclosure, 26).

CHARGER



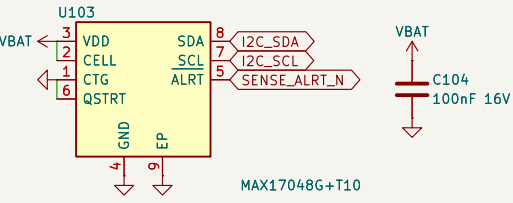
BQ24073 linear charger with power path (DPPM): runs MAO from USB while charging. USB500 input limit (EN2=0, EN1=+3V3), ISET 4.3k = 207 mA charge (cell maximum 250 mA = 0.35), ILIM 1.5k, default timers. Pack NTC on TS; the charger stops outside 0–50 C; the cell allows 0–40 C, so firmware pauses charging through /CE (expander P6) above 43 C board temperature; charging state comes from the gauge (CRATE) and PGOOD, so /CHG is not used. On F under the panel: its heat stays off the cell.

BATTERY



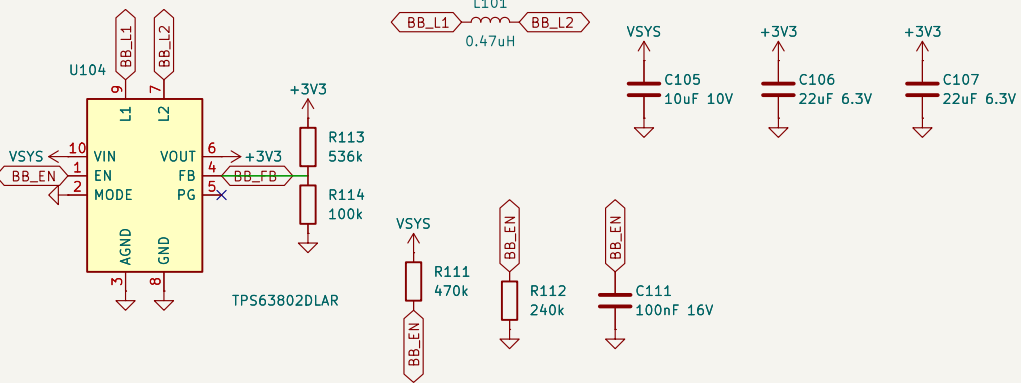
JST SH 3-pin: 1 = BAT-, 2 = NTC, 3 = BAT+ (same order as KINO J1100, ODD JOBS 41/167). Cell MUST carry its own protection PCM and a 10k NTC. R108 = 0R link to measure battery current (110). Q101 blocks a reversed cell (51); with USB in, the charger stays in its 4–11 mA short probe. R110 fits only for cells without NTC.

FUEL GAUGE



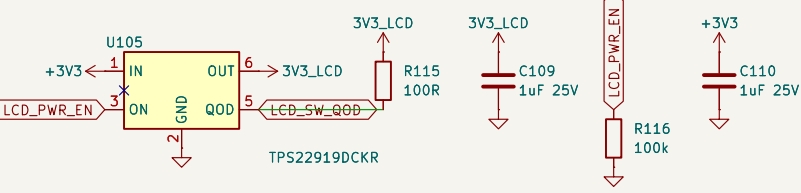
MAX17048 ModelGauge on VBAT, I2C 0x36, ALERT wired-OR with the light sensor INT.

3V3 BUCK-BOOST



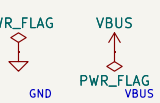
TPS63802: VSYS 2.9–4.4 V to +3V3, 2 A, 11 uA Iq. EN divider = hardware UVLO: on at 3.25 V, off at 2.96 V (protects the cell even if firmware fails), 100 nF so load steps cannot trip it. MODE low = PFM. FB 536k/100k = 3.18 V (3.10–3.28 V worst case; under the panel VCI 3.3 V maximum). L = 0.47 uH only.

DISPLAY RAIL

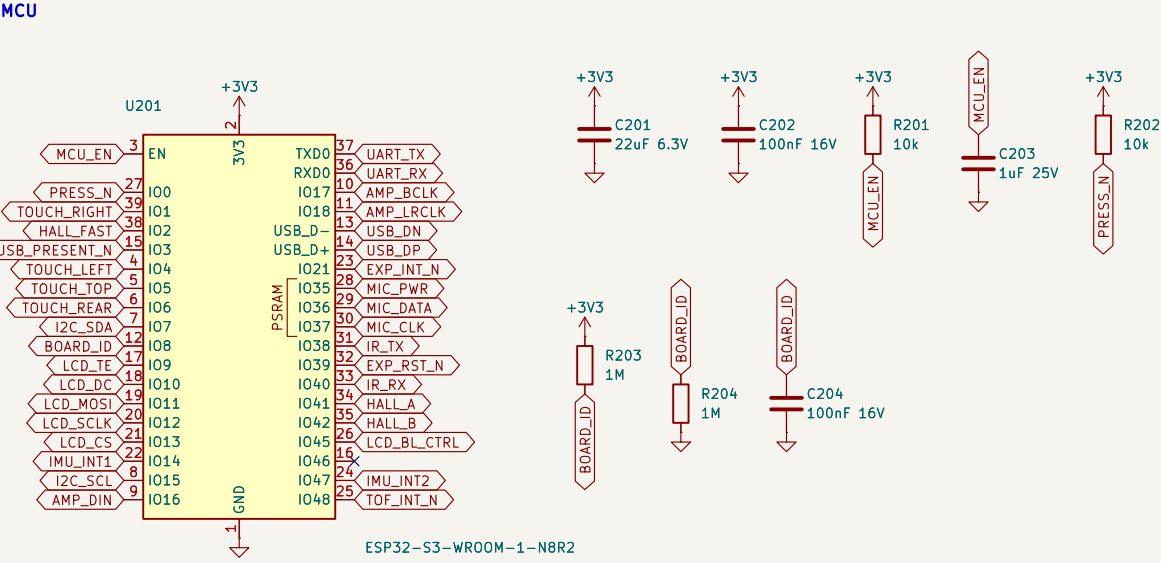


TPS22919 switches 3V3_LCD (panel logic; the backlight runs from VSYS). ON pulled down; off until firmware enables it. Fixed 1.6 ms soft start (datasheet 1.75 ms at 3.6 V), so the -20 mA panel inrush cannot brown out the MCU. QOD to VOUT through 100R; clean power-down. SC70–6, 1.1 mm: it sits under the panel (zone A <= 1.2 mm).

RAIL SOURCES

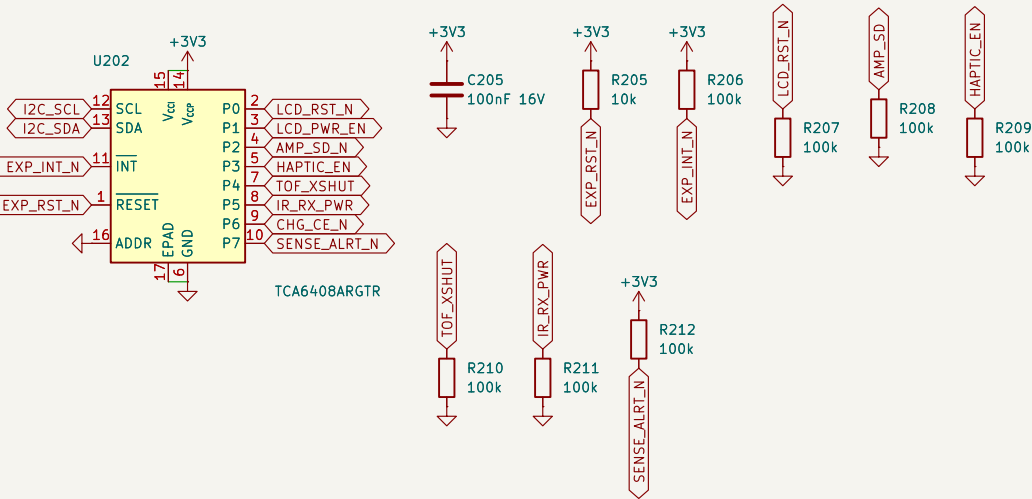


COMPUTE



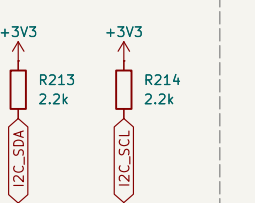
ESP32-S3-WROOM-1-NBR2 on B.Cu at 6 o'clock, antenna over the board-edge notch (no copper beneath). 22 uF + 100 nF at 3V3. EN: 10k/1 uF RC. GPIO0 = face press (BOOT). Unused pins NC or test pads.

EXPANDER



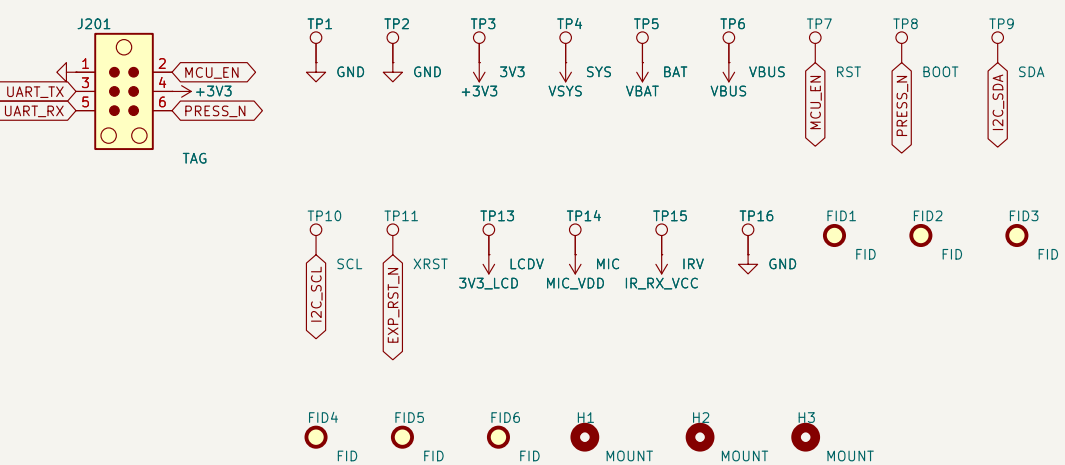
TCA6408A at 0x20. Outputs default off through 100k pull-downs; inputs pulled up. Firmware writes the output register before switching pins to outputs (register resets to 0xFF).

I2C



One 400 kHz bus. 2.2k pull-ups (ODD JOBS 30); -6 devices, short traces.

SERVICE



Tag-Connect TC2030-NL (no part fitted): 1 GND, 2 EN, 3 TXD0, 4 3V3, 5 RXD0, 6 GPIO0 - recovery without USB (ODD JOBS 33-35, 157). Pinout chosen so TXD0/RXD0 run straight from the module pins. Probe pads: rails beside the charger, comms in one field, switched rails at their sources (36-39).

ESP32-S3 module, GPIO expander, boot/reset, service and test access

ODD JOBS

Sheet: /Compute/
File: compute.kicad_sch

Title: MAO_MAIN A0 - Compute

Size: A3
KiCad E.D.A. 10.0.6

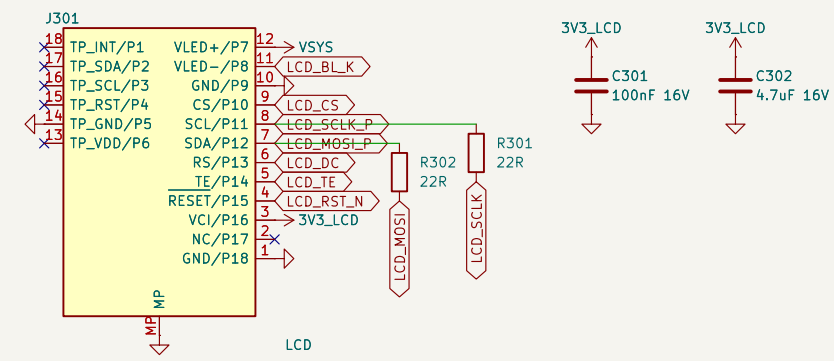
Date: 2026-10-04

Rev: A0

Id: 3/6

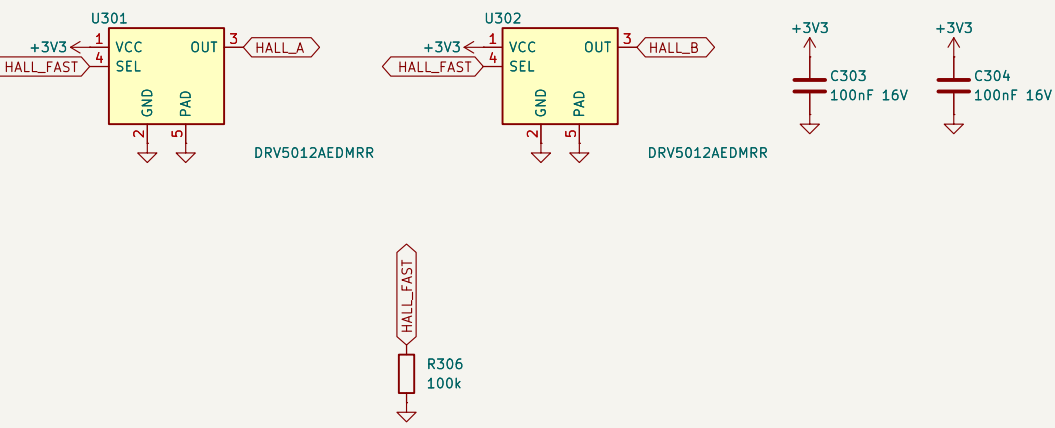
INTERFACE

DISPLAY



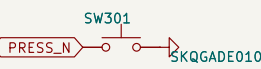
Plug-in display: 1.28" round GC9A01 panel (Winstar WF0128BTYAA4DNN0, 240x240 IPS) on an 18-pin 0.5 mm FPC tail, folded under the panel into a 1.0 mm high back-flip connector with top and bottom contacts, so any panel of this 18-pin standard plugs in either way up and can be swapped without solder. Pins 1–6 serve the touch controller of touch variants; not wired (MAO's face is a pressed window). TP_GND to GND. TE to GPIO9 for tear-free frames. 22R on SCLK/MOSI damps 80 MHz edges (ODD JOB5 29). Panel logic on the switched 3V3_LCD rail, backlight anode on VSYS through the constant-current driver; RST held low until firmware. The stock 70 mm tail 5-folds in the face carrier, drops through the board slot at 9 o'clock and plugs into J301 on B.Cu.

RING DIAL



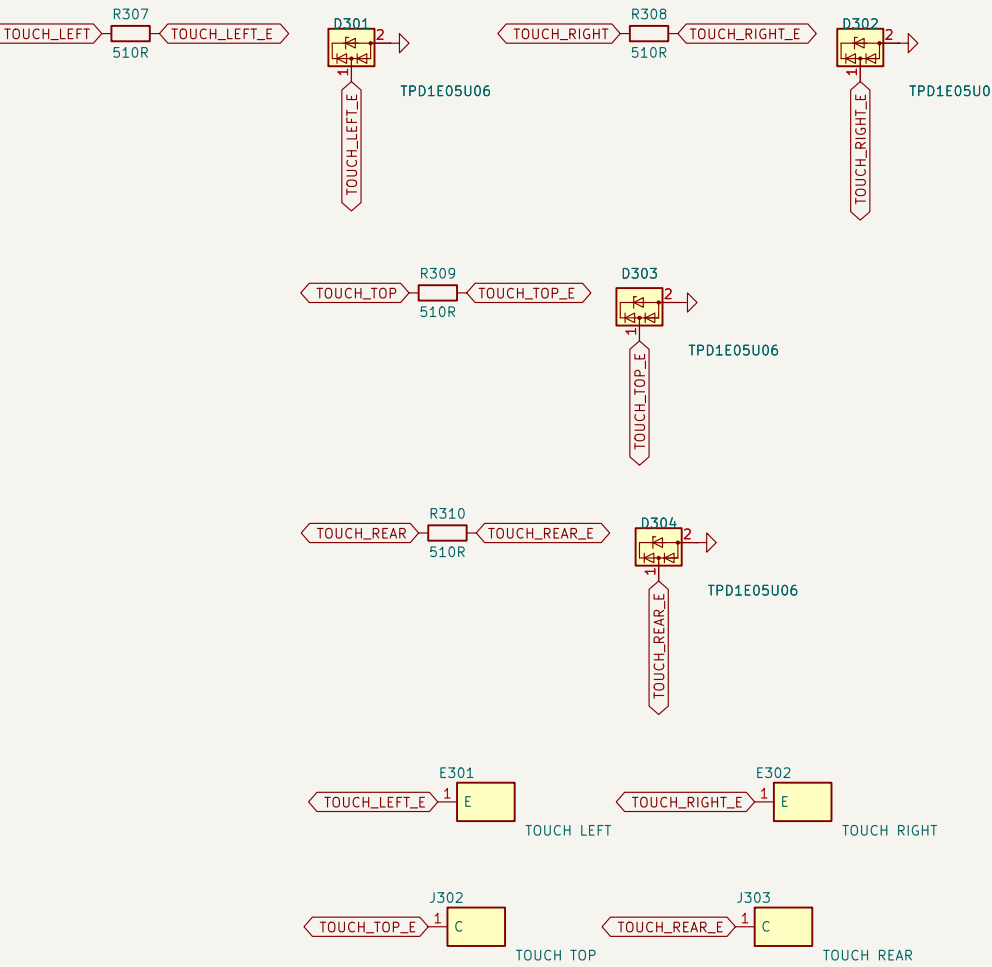
Two DRV5012 Hall latches under the ring's 30-pole ferrite strip, 6 deg apart (half a pole = 90 deg electrical): A/B quadrature identical to the LCDKit EC11 (2 transitions per detent, 30 detents). HALL_FAST low = 20 Hz sampling, 1.6 uA (sleep); high = 2.5 kHz (awake).

FACE PRESS



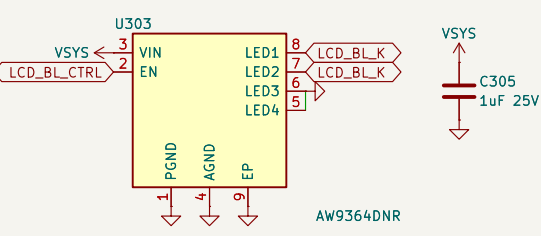
ALPS SKQG on F.Cu at the centre, under the panel; the floating face presses it. GPIO0 = BOOT strap: no capacitor here.

BODY TOUCH



Four ESP32–S3 touch channels. 510R within 1 mm of the module (Espressif guide); 0.42 pF ESD at the electrode. LEFT/RIGHT: copper arcs at the rim; TOP/REAR: gold springs to printed or foil electrodes.

BACKLIGHT



Panel LEDs (one group of two) need VLED+ 3.0/3.2/3.4 V at 40 mA (Winstar spec 4.2); more than 3V3 can drive. AW9364 constant-current sinks from VSYS (40–50 mV dropout): LED1+LED2 on VLED– = 2 x 20 mA = 40 mA nominal (33–47 mA; +-17.5 % part tolerance); set by the part, not by firmware. 16 steps by EN pulses on GPIO45; its 150k EN pull-down keeps the panel dark from Pset (ODD JOB5 116). Full 40 mA down to VSYS 3.4 V on the typical 3.2 V bin, 3.6 V on the 3.4 V bin (sim S3); dimmer below.

Round display, backlight, ring dial, face press, body touch

ODD JOBS

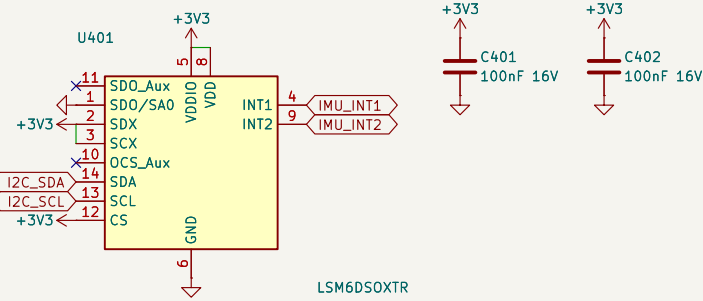
Sheet: /Interface/
File: interface.kicad_sch

Title: MAO_MAIN A0 – Interface

Size: A3	Date: 2026–10–04	Rev: A0
KiCad E.D.A. 10.0.6		Id: 4/6

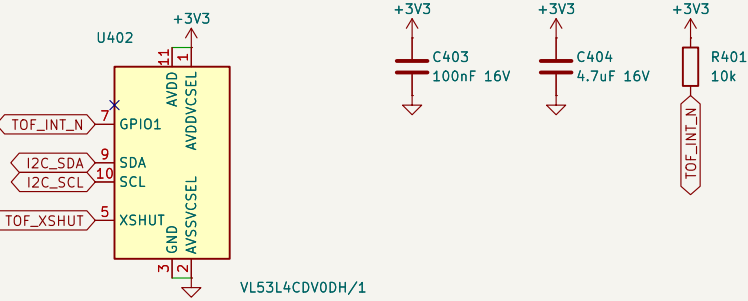
SENSE

IMU



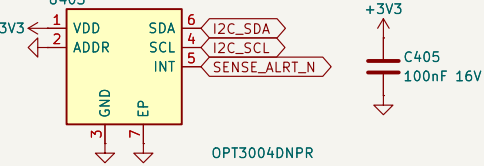
LSM6DSOX at 0x6A (SA0 low), I2C (CS high), SDx/SCx to VDDIO keeps the land compatible with the LSM6DS3TR-C fallback. INT1/INT2 push-pull, configured active-low for ext1 wake.

PROXIMITY



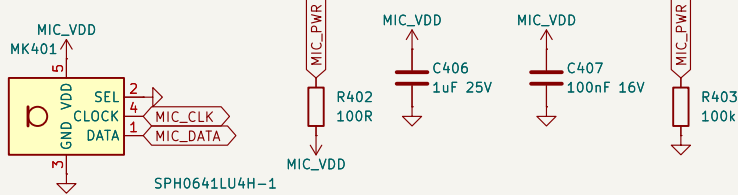
VL53L4CD time-of-flight, 0–1.3 m, under the window border. XSHUT from the expander (gff by default); GPIO1 open-drain with the 10k pull-up ST recommends. AVDD decoupled 4.7 uF + 100 nF per ST.

AMBIENT LIGHT



OPT3004 at 0x44, human-eye response, under the window border. INT joins SENSE_ALRT_N.

MICROPHONE



SPH0641 bottom-port PDM mic on B.Cu, port through the board (0.5 mm hole) to the top. Powered from GPIO35 through 100R/1 uF only while listening; it draws 80 uA even with the clock stopped. SEL low.

RAIL SOURCES



IMU, proximity, ambient light, microphone

ODD JOBS

Sheet: /Sense/
File: sense.kicad_sch

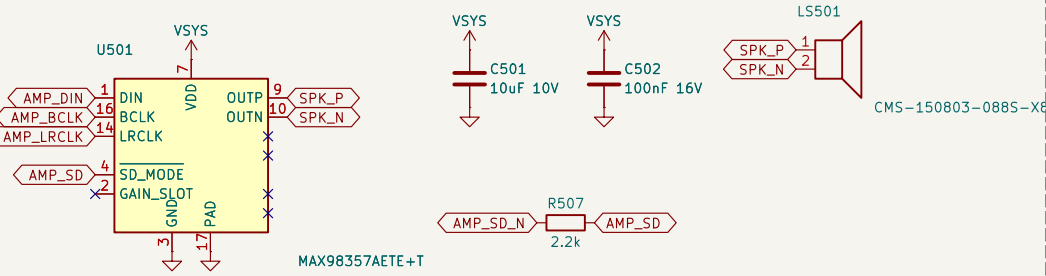
Title: MAO_MAIN A0 – Sense

Size: A3 Date: 2026-10-04
KiCad E.D.A. 10.0.6

Rev: A0
Id: 5/6

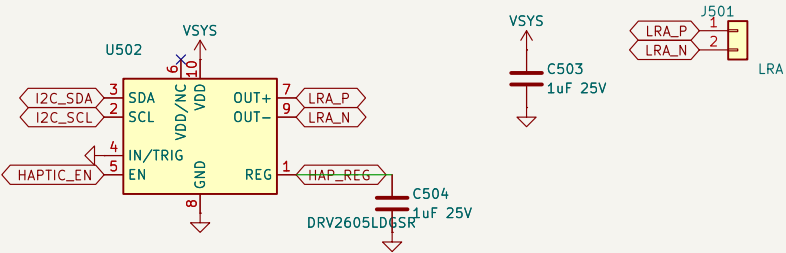
FEEDBACK

SPEAKER AMP



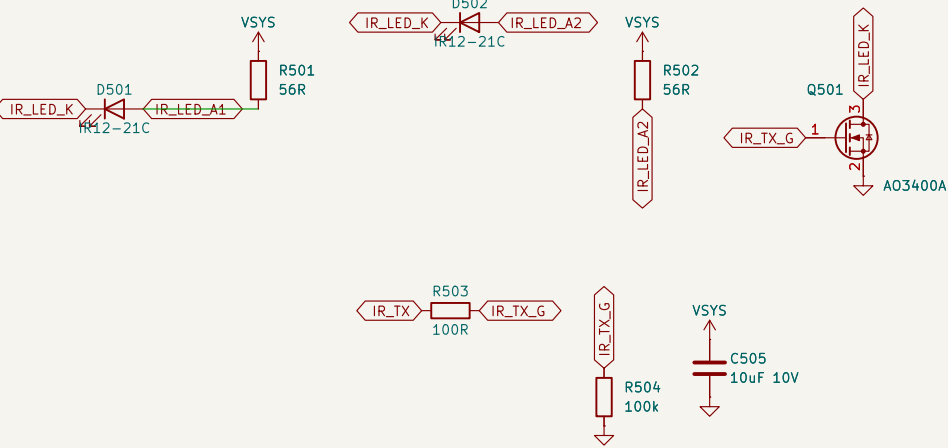
MAX98357A I2S class-D from VSYS. SD_MODE from the expander through 2.2k (datasheet: series R when VDDIO can exceed VDD); lqw = 0.6 uA shutdown, 3.2 V = left channel, GAIN_SLOT open = 9 dB (datasheet state, deliberate). Speaker: Same Sky CMS-150803-088S-X8 (15 x 8 x 3 mm, 8 ohm, 0.8 W) in the base at 9 o'clock; its own spring contacts press on two pads on B.Cu (LS501).

HAPTIC



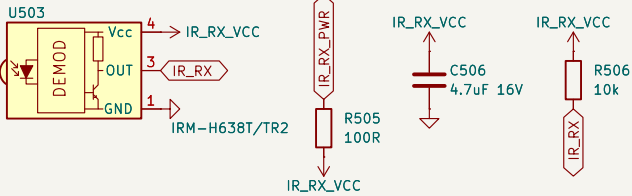
DRV2605L closed-loop LRA driver from VSYS. I2C 0x5A, EN from the expander (4 uA off), IN/TRIG low: playback by I2C. Pin 6 VDD/NC left open (SLOS854D: optional supply, tie to VDD or float). LRA LD0832AA (235 Hz, 1.8 Vrms) glued to the base, leads to J501.

IR TRANSMIT



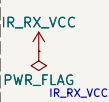
Two side-emitting 940 nm LEDs at the back edge (B.Cu), each with its own 56R from VSYS (26-59 mA pulses at 38 kHz, 33 % duty; under the 65 mA rating at any VSYS up to 4.5 V and any Vf). AO3400A low side, gate pulled down and driven from GPIO38, a pin without a reset pull-up: dark at reset (117).

IR RECEIVE



IRM-H638T (as LCDkit), top-view, looking up through the window border. Powered from expander P5 through 100R/4.7 uF only when listening (0.4 mA). OUT pull-up goes to the switched supply, never to +3V3, so an unpowered receiver is never back-powered and GPIO40 reads a defined low while it is off.

RAIL SOURCES



Speaker amplifier, haptic driver, IR transmit and receive

ODD JOBS

Sheet: /Feedback/
File: feedback.kicad_sch

Title: MAO_MAIN A0 - Feedback

Size: A3 Date: 2026-10-04

KiCad E.D.A. 10.0.6

Rev: A0

Id: 6/6